

Switching Concepts — Guided Notes

Unit 2.1.1 | CompTIA Network+ N10-009 Obj. 2.1 | N10-008 2.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A switch learns MAC addresses by reading the _____ on every incoming frame and recording which _____ it arrived on. Entries age out after typically _____ seconds. This table is also called the _____ table.

Question 2

Match each switching behavior to its condition:

Forwarding → _____

Filtering → _____

Flooding → _____

Question 3

Each switch port is its own _____ domain, eliminating collisions. By default, the entire switch is one _____ domain — a broadcast reaches every port. _____ and _____ are the two tools used to break broadcast domains.

Question 4

Full duplex allows a device to _____ and _____ simultaneously, effectively doubling available bandwidth. A duplex mismatch occurs when one side negotiates _____ and the other negotiates _____. The symptom is _____.

Question 5

Without STP, redundant switch links create _____. A broadcast frame enters the loop and circulates _____, multiplying at every junction. This is called a _____ and will take down the network in seconds.

Question 6

In STP, the switch with the lowest _____ becomes the root bridge. The Bridge ID is composed of _____ and _____. In production, you should set _____ manually to control which switch wins the election.

Question 7

A port in STP _____ state receives BPDUs but does not forward frames. The transition from blocking to forwarding in classic STP takes approximately _____ seconds. RSTP reduces this to _____ and consolidates port states to: _____, _____, and _____.

Question 8 — Real World Application

REAL WORLD: A user plugs a cheap unmanaged switch into a wall jack to add more ports at their desk. Within seconds, the entire floor loses network connectivity. Explain what happened and what configuration would have prevented it.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-1-1-switching-concepts/2-1-1-switching-concepts-answer-key.pdf
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